

Agentic AI System: Child YouTube Safety Monitor

Problem Statement & Motivation

In this new technological age, tablets are a device that most parents are giving their toddlers and small children to distract them while they manage adult responsibilities such as work and household management, being on top of what their children are viewing on their tablets can be challenging. While there are parental controls that can be placed on these devices, they rely on binary rules which can fail to detect contextually inappropriate content or unnecessarily block educational material.

Use Cases & Scenarios

Case 1:

- A child is watching what appears to be a child-friendly cartoon on YouTube. The video is brightly colored, animated, and not age-restricted, so the standard parental controls allowed the video to play. However, in the video the dialogue includes sexual innuendos and suggestive humor that is not needed for children.
 - Agent will...
 - Detect the video being played
 - Convert dialogue to text using speech to text
 - Identifies repeated double meanings, sexual innuendo phrasing, and adult humor
 - Sends an alert to parent/guardian through text and/or email

Case 2

- A child is trying to watch a YouTube video for their World War I project for school. The video is blocked due to violence and weapons being shown and mentioned
 - Agent will...
 - Detect the video being shown
 - Pulls metadata of what channel is categorized as, description referencing learning objectives, and check if channel has prior violations
 - Identifies transcript and visuals for appropriateness of content
 - Allows playback
 - Notifies parent/guardian of video title, video content, and reason for override

Initial System Design

Agents

- Activity Monitoring Agent
 - Detects video being played, collects timestamps and video ID
- Content Retrieval Agent
 - Pulls metadata
 - Converts dialogue to text using speech-to-text
 - Analyzes language patterns and phrasing
 - Captures repeated double meanings, inappropriate phrases and actions
 - Humor that is not child appropriate
- Real-Time Intervention Agent
 - Pauses video or allows playback of videos that it is

- Displays a message that shows child that content is not appropriate
- Alert/Notification Agent
 - Alerts parent/guardian of what they found and reason for blocking or allowing content
- Learning/Feedback Agent
 - Parent/guardian confirms decision
 - System learns to reduce false positives and false negatives for future content

Data Source(s): YouTube

Feasibility & Risks

- False negatives/false positives
- Limited access to YouTube data
- Multimodal analysis can be computationally expensive

Timeline

02/05 – 03/04: Build interface for AI Agent

03/04-03/11: Create multi-agent workflow

03/12-03/25: Evaluate the model through testing

03/26-04/22: Practice presenting agent